Sahaj Bhadja

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Summary

Full-stack Software Engineer pursuing an MSCS at USC with 2+ years building **LLM-powered products** with **TypeScript/React** front-ends and **Python/FastAPI** microservices on GCP, spanning SQL/NoSQL/blob storage, production operations, and Generative AI tooling.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Database Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **fault-tolerant LLM orchestration** with FastAPI microservices and async job-state handling across 5K+ monthly jobs, scaling horizontally with explicit failure isolation at 99%+ production reliability.
- Designed **multi-tenant data storage** on SQL and blob storage for 500+ users, optimizing schemas, indexing, and queries to cut latency from 1.2s to 350ms across production workloads.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **full-stack LLM features** with React/TypeScript and Python (Flask) microservices integrating multimodal models behind a switchable architecture, serving 10K+ daily requests in production.
- Wrangled **Generative AI tools (Cursor, Claude)** and automated testing/CI-CD to multiply productivity, shipping reliable LLM-powered features for 150+ users in a fast-paced startup environment.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built and deployed **production-grade ML model pipelines** over 100K+ records with transformer architectures, improving evaluation **18%** through rigorous benchmarking and experimentation.
- Engineered **reproducible training and evaluation infrastructure** (PyTorch, config-driven), enabling reliable model iteration and high-confidence deployment decisions across large-scale experiments.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Built **Python APIs and microservices** with OOP design and PyTest-based testing on AWS (Glue, S3), improving response times **30%** across distributed production systems.
- Integrated **CI/CD pipelines and 20+ automated test suites** (Jenkins, PyTest), increasing reliability **68%** through code reviews, monitoring, and configuration-managed deployments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **full-stack applications** (React, Node.js) with microservices and real-time updates across eight workflows, syncing 2M+ records with SQL and NoSQL data storage.
- Automated **Docker-based deployment and monitoring** via Jenkins with Sentry and Azure Monitor, cutting release time **50%** and improving production stability and observability.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, Bash
AI & LLMs: LLM Orchestration, Generative AI, Multimodal Models, Transformers, PyTorch, RAG, AI Code Gen (Cursor, Claude)
Backend & APIs: FastAPI, Pydantic, Flask, Node.js, REST APIs, Microservices, uv/ruff
Frontend: React.js, Next.js, TypeScript, TailwindCSS, Component-Based UI
Data Storage: PostgreSQL, MySQL, MongoDB (NoSQL), SQL Server, Blob Storage, Firebase
Cloud & DevOps: GCP, AWS, Azure, Docker, Kubernetes, Terraform, CI/CD, Monitoring

Projects

Big-Brain Live | *GCP, LLMs, FAISS, Pinecone*
- Built an **AI document platform on GCP** with LLM-powered semantic search and summarization over 50K+ records using FAISS and Pinecone, reducing analysis time **60%** with sub-second retrieval.

PiXMod Live | *Next.js, Transformers, MongoDB, Stripe*
- Built an **AI-powered Next.js SaaS platform** integrating transformer models (CLIP, SAM, Stable Diffusion) with MongoDB storage and Stripe billing, processing 1K+ assets with scalable design.

Chat2SQL Code | *Java, TypeScript, SQL, NoSQL*
- Built an **AI-powered SQL workspace** generating schema-aware queries across 10+ SQL and NoSQL databases in a Java and TypeScript codebase with Docker-backed deployment.